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Device-Aware Threshold Personalization: A Controlled Threshold-Calibration Study for Non-IID Federated IoT Anomaly Detection

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***Abstract*—Objective: This study isolates threshold calibration scope as the sole experimental variable in a federated IoT anomaly-detection pipeline. A fixed FedAvg autoencoder ($E{=}1$, full participation) is trained once per seed; the same encoder, seeds, and per-client score artifacts are reused across shared (B1), per-client (B2), family-mean (B3), and cluster-mean (B4) threshold policies—a threshold-only controlled comparison. Protocol: The primary comparison is B1 vs. B2 on N-BaIoT ($K{=}9$ physical devices, $q{=}0.95$, five seeds). Because B2 sets each client's threshold at its own benign p_{95}, FPR equalization is expected by construction under stable calibration/test distributions; the empirical contribution is the held-out magnitude, seed consistency, detection-quality tradeoff, and B4's taxonomy-free approximation. CICIoT2023 (Regime B) uses randomized file-level pseudo-clients—not physical devices—as a near-homogeneous applicability boundary. Results: B2 reduces CV(FPR) from 1.017 to 0.299 ($\Delta{=}0.718$, bootstrap CI $[0.647, 0.769]$, all five seed deltas positive). B4 recovers $\approx 52\%$ without a taxonomy. The gain is not uniform: lower-tail Macro-F1 degradation concentrates in a single low-separability client. Under the CICIoT2023 partition, no improvement is observed. Limits: Nine devices; $E{=}1$ limits local drift; five seeds; CICIoT2023 conclusions hold only under the tested randomized partition.**

***Index Terms*—federated learning, IoT anomaly detection, threshold calibration, false-positive-rate dispersion, autoencoder, non-IID data, personalization.**

I. Introduction

Federated learning (FL) trains shared models without pooling raw traffic [1]. Reconstruction-error autoencoders (AEs) are a natural fit for IoT intrusion detection because benign-only training suffices and attack labels are not required at training time.

A standard deployment trains one shared AE via FedAvg, then applies a threshold τ above which a reconstruction error signals an intrusion. In homogeneous fleets this works well. In heterogeneous fleets, however, benign reconstruction-error distributions differ substantially across device types: a single shared τ imposes unequal false-positive rates (FPRs). Some devices carry a disproportionate false-alarm burden—making them noisy or operationally unusable—while others may become under-sensitive. A shared server-broadcast threshold is natural in deployments where the system exposes a single global alarm boundary or where per-client threshold state is not managed independently.

This study asks how large the per-client FPR disparity is under a shared threshold, how much per-client calibration reduces it, and what detection-quality tradeoff this creates—all under a fixed FedAvg AE with real physical-device heterogeneity. The target operational scenario is a fleet of physical IoT devices for which local benign calibration data are available, and where reducing unequal false-alarm burden across devices is a deployment priority. Prior FL-AE work may report aggregate detection metrics or use thresholds implicitly, but without fixing the encoder, aggregation protocol, and training hyperparameters and varying only threshold scope, it is impossible to attribute per-client alarm-rate dispersion to the threshold policy rather than to training differences. This controlled-comparison gap motivates the present study.

The primary metric is the coefficient of variation of FPR across eligible clients, CV(FPR) $= \sigma_{\text{FPR}}/\mu_{\text{FPR}}$. Four threshold policies are evaluated: a client-averaged shared threshold (B1), full per-client personalization (B2), device-family grouping (B3), and unsupervised cluster grouping (B4). B1 is constructed as the arithmetic mean of eligible clients' local 95th-percentile benign calibration thresholds, broadcast as a single scalar. The arithmetic mean treats eligible clients equally and models a simple server-broadcast threshold derived from local calibration summaries, avoiding sample-count domination; pooled and sample-weighted alternatives are reported as sensitivity checks. A centralized baseline (B0) is included as a privacy-incompatible reference comparator but is not part of the single-variable B1–B4 comparison.

The study uses N-BaIoT [2] as the primary confirmatory regime (Regime A), CICIoT2023 [3] as a file-level applicability check under near-homogeneous conditions (Regime B), and synthetic Dirichlet partitions of N-BaIoT as a heterogeneity severity sweep (Regime C). Five random seeds are used; seed-level descriptive summaries (mean, SD, range, sign consistency) report stability. The FL simulation uses Flower [4].

This article contributes a threshold-only calibration analysis for federated IoT anomaly detection. It quantifies the false-positive-rate disparity induced by shared thresholds across heterogeneous physical devices, evaluates per-client, family-mean, and cluster-mean calibration under a fixed FedAvg autoencoder, characterizes the resulting detection-quality tradeoff, and provides reproducible source code, configurations, per-seed results, and figure-generation scripts through an archived replication artifact.

The remainder of this article is organized as follows. Section II reviews related FL autoencoder, non-IID aggregation, model-personalization, and threshold-personalization work; Sections III–V define the alarm decision, dispersion metric, threshold-personalization framework, datasets, regimes, and evaluation protocol; Sections VI–VIII report the results, discuss validity threats, and conclude with practitioner guidance, limitations, and future work.

II. Related Work

Federated autoencoders for IoT anomaly detection have been applied to device-partitioned N-BaIoT by Rey et al. [5], Khraisat et al. [6], and Olanrewaju et al. [7], who report aggregate detection metrics. Thresholds are applied implicitly or globally; none isolates threshold calibration scope as a variable or measures per-client FPR dispersion.

Non-IID aggregation work such as FedMSE [8] improves AUC under Dirichlet non-IID N-BaIoT by changing aggregation weights, not post-training threshold policy; per-client alarm-rate dispersion is not reported [9], [10].

Model-level FL personalization methods such as Per-FedAvg, FedPer, and LG-FedAvg [11]–[13] adapt weights or training objectives, whereas DATP fixes the shared AE and adapts only the post-training alarm boundary.

Threshold-level personalization is acknowledged in Kitsune [14] and DIoT [15], but neither provides a controlled comparison fixing encoder, training protocol, and seeds across threshold-scope variants. Recent FL-AE IoT IDS and non-IID aggregation studies improve aggregate detection or training behavior, but threshold scope remains implicit, fixed, or unvaried; conversely, local-threshold systems acknowledge per-device calibration without isolating it under a frozen federated encoder. Table I summarizes this gap.

III. Problem Formulation

A. Notation

Let $\mathcal{K} = \{1, \ldots, K\}$ denote the set of FL clients. Each client k holds local benign traffic data $\mathcal{D}_k^{\text{ben}}$ and may have attack data $\mathcal{D}_k^{\text{atk}}$. An autoencoder $h = g \circ f$ maps input $\mathbf{x} \in \mathbb{R}^d$ to a reconstruction $\hat{\mathbf{x}} = g(f(\mathbf{x}))$, where f is the encoder and g the decoder. The reconstruction error is the per-sample mean squared error:

$$e(\mathbf{x}) = \frac{1}{d}\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2, \tag{1}$$

TABLE I
POSITIONING: ADAPTATION LAYER AND THRESHOLD TREATMENT.

Work group	Adaptation layer	Threshold treatment
FL-AE IoT IDS [5], [6]	Model (shared AE)	Implicit/global
FedMSE [8]	Aggregation weights	Not varied
Per-FedAvg, FedPer [11], [12]	Model weights/ objectives	Not addressed
Kitsune, DIoT [14], [15]	Local AE	Per-device (no controlled study)
DATP (this work)	**Threshold only**	**Controlled comparison**

where d is the input dimension.

B. Binary Alarm Decision

Given a threshold τ_k for client k, the alarm decision is:

$$\text{anomaly}(\mathbf{x}) = \mathbf{1}[e(\mathbf{x}) > \tau_k]. \tag{2}$$

The FPR for client k is $\rho_k = \Pr[e(\mathbf{x}) > \tau_k \mid \mathbf{x} \sim \mathcal{D}_k^{\text{ben}}]$.

C. Dispersion Metric

Let $\mathcal{K}_{\text{elig}} \subseteq \mathcal{K}$ denote clients with at least $n_{\min} = 100$ benign calibration samples; other clients are calibration-pending and excluded from dispersion metrics. The primary metric is the coefficient of variation of FPR over $\mathcal{K}_{\text{elig}}$:

$$\text{CV(FPR)} = \frac{\sigma_\rho}{\mu_\rho}, \quad \mu_\rho = \frac{1}{|\mathcal{K}_{\text{elig}}|} \sum_{k \in \mathcal{K}_{\text{elig}}} \rho_k, \tag{3}$$

where σ_ρ is the standard deviation. CV(FPR) normalizes cross-client FPR dispersion by the fleet mean, making dispersion comparable across regimes with different mean FPR levels; IQR(FPR) and max–min FPR are reported as absolute-dispersion checks to guard against small-denominator artifacts.

Eligibility and fallback. Calibration-pending clients receive the client-averaged shared threshold τ_{global} (B1 value) as a fallback. Coverage is reported as $|\mathcal{K}_{\text{elig}}|/|\mathcal{K}|$ for all conditions.

Operational assumption. Percentile calibration assumes each client's benign reconstruction-error distribution is sufficiently stable between calibration and test. Temporal drift would weaken the calibration guarantee; this paper does not model concept drift.

D. Confirmatory Statistic

The primary statistical estimate is the seed-level delta:

$$\Delta_s = \text{CV(FPR)}_{\text{B1},s} - \text{CV(FPR)}_{\text{B2},s}, \tag{4}$$

aggregated over seeds $s \in \mathcal{S}$ via descriptive summaries: mean, standard deviation, range, and sign consistency. With only five seeds, these statistics are reported as a seed-level consistency summary—not a high-powered population significance test. All-positive seed deltas are interpreted as seed-level support for RQ2 in this tested protocol.

E. Research Questions

- **RQ1:** Under the client-averaged shared threshold (B1), how large is the variation of FPRs across heterogeneous IoT clients?
- **RQ2:** What held-out FPR-dispersion reduction does per-client threshold calibration (B2) achieve relative to the client-averaged shared threshold under physical-device heterogeneity, what TPR/Macro-F1 tradeoff does this impose, and how much of the per-client benefit does taxonomy-free cluster grouping (B4) recover?
- **RQ3:** How much of the B2 dispersion reduction can cluster-mean calibration (B4) recover?

The benign calibration-error distributions of different device types differ substantially in location and spread, so the B1 client-averaged threshold τ_{global} crosses each distribution at a

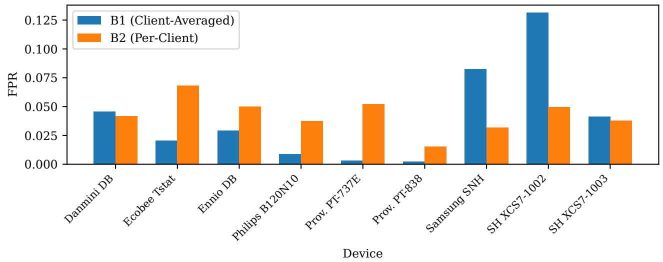

Fig. 1. Per-device FPR under B1 (Client-Averaged Shared τ) and B2 (Per-Client τ) for the nine N-BaIoT clients. *Illustrative only (seed 0); multi-seed inference uses Table IV and Table VI.*

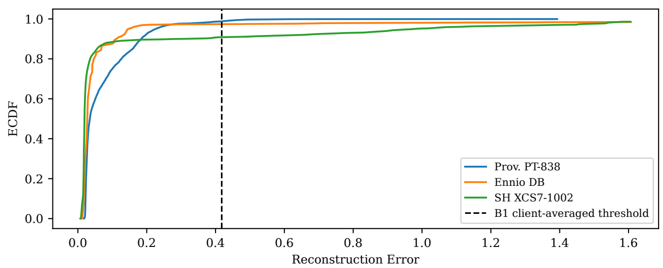

Fig. 2. Benign reconstruction-error empirical cumulative distributions for three representative N-BaIoT clients (seed 0). The shared threshold τ_{global} (B1, dashed) crosses each curve at a different exceedance fraction $(1-F(\tau))$, producing unequal FPRs. Per-client thresholds (B2) set each client's p_{95} independently, aligning benign-tail exceedance rates by construction. Attack distributions are not shown; they need not shift in parallel with benign tails, explaining the TPR/Macro-F1 tradeoff observed in Table IV.

different fraction, producing device-specific FPRs. Because B2 uses each client's local benign percentile, FPR equalization is expected when calibration and test distributions are stable; the empirical questions are therefore the held-out magnitude of this expected calibration effect, its seed-level consistency, the detection-quality cost it creates, and the extent to which grouped calibration (B4) approximates per-client calibration without a device taxonomy—all under a fixed FL-AE setting with real physical-device heterogeneity.

IV. Threshold Personalization Framework

Fig. 1 illustrates the operational consequence of heterogeneous calibration: under B1 (client-averaged shared τ), per-device FPRs vary substantially across the nine N-BaIoT clients, while B2 (per-client τ) reduces this dispersion for the representative seed shown. As an illustrative preview of the primary Regime A pattern, Fig. 1 shows seed 0 only; inference uses the multi-seed results in Table IV.

Pipeline. Each client holds benign traffic split into train, calibration, and test partitions; attack data are reserved exclusively for evaluation and are never used to fit or tune thresholds. A shared AE is trained with FedAvg via the Flower framework [4] (identical training across B1–B4; B0 uses centralized pooling). *The AE is trained once per (dataset, regime, seed, α) condition.* After training, each client scores its calibration and test data under the fixed AE; per-client score arrays are written to disk and reused by all threshold policies without retraining. Even when a run is budget-capped at 150 rounds, the same final AE state is shared across all B1–B4 derivations, preserving the threshold-scope-only comparison.

Threshold policies. Four policies operate on the shared score artifacts:

- **B1 (Client-Averaged):** $\tau_{\text{global}} = \frac{1}{|\mathcal{K}_{\text{elig}}|}\sum_{k\in\mathcal{K}_{\text{elig}}}\tau_k^{(95)}$, applied uniformly to all clients. Pooled and weighted variants are sensitivity checks only (Table VII).
- **B2 (Per-Client):** $\tau_k = \tau_k^{(95)}$ applied locally; zero additional uplink communication in a deployment where each client calibrates its threshold locally.
- **B3 (Family-Mean):** arithmetic mean of local thresholds within each device family (Regime A only; requires a device taxonomy).
- **B4 (Cluster-Mean):** k-means on a 4-scalar fingerprint $[\mu_e, \sigma_e, \text{skew}_e, p_{95}(e)]$ (StandardScaler-normalized); $K{=}3$ for Regime A, silhouette-selected for Regimes B/C. For silhouette selection (Regimes B/C), candidate $K \in \{2,3,4,5\}$; k-means uses `k-means++` initialization, `n_init`=10, `max_iter`=300, `random_state`=42, applied after StandardScaler normalization of the fingerprint matrix. The four statistics are compact descriptors of the benign calibration-error distribution: μ_e captures location, σ_e spread, skew_e tail asymmetry, and $p_{95}(e)$ the operating tail used by thresholding. They are distributional summaries for grouping, not privacy mechanisms. In Regime A, $K{=}3$ is a practical fixed choice: with only 9 physical devices and coarse device-family structure, silhouette-based K selection is unstable at small n; $K{=}3$ avoids overfitting cluster count to sampling noise. B4 requires sharing these distributional fingerprints, which carries nonzero metadata leakage risk (Section VII).

Clients with fewer than $n_{\min}{=}100$ benign calibration samples are calibration-pending; they receive τ_{global} as fallback and are excluded from CV(FPR). **Evaluation.** Each client evaluates FPR, TPR, balanced accuracy, and Macro-F1 on held-out test sets. CV(FPR), CV(TPR), IQR(FPR), worst-client balanced accuracy, P_{10} Macro-F1 (10th percentile of per-client Macro-F1 across eligible clients), and coverage ratio are aggregated. Seed-level deltas are summarized descriptively (mean, SD, range, sign consistency).

V. Experimental Design

A. Datasets and Regimes

Regime A — N-BaIoT (confirmatory). The N-BaIoT dataset [2] contains traffic from nine physical IoT devices grouped by the configured device-family taxonomy, infected with Mirai and BASHLITE botnets. Each physical device is one FL client ($K = 9$). The N-BaIoT natural partition mixes device-specific benign feature heterogeneity with some attack-variant skew, and is therefore treated as a realistic natural partition rather than a pure feature-skew isolation regime. Regime A is the primary planned condition for RQ2.

Regime B — CICIoT2023 (file-level applicability check). CICIoT2023 [3] provides 63 file-defined pseudo-clients under the partition used here. Regime B uses randomized file-level pseudo-clients, not physical IoT devices. Each of the 63 file-defined clients corresponds to one merged flow-level traffic file from the dataset's `MERGED_CSV` directory; client boundaries are defined by the dataset's own file-level granularity, not by physical device identity. Within each client, benign records are randomly shuffled (no temporal ordering is preserved) and split 70/15/15 into train, calibration, and test-benign partitions; attack records are reserved exclusively for test-attack evaluation. A per-client cap of 50,000 total records is applied, reserving up to 10,000 attack records; this bounds computation and balances benign/attack evaluation volume while preserving the fixed controlled comparison (after capping, the effective per-client evaluation mix is approximately 20% benign and 80% attack, with median benign count ≈2,500 per client; exact counts are recorded in the replication artifact). Regime B tests whether the Regime A personalization benefit holds when benign reconstruction-error distributions are near-homogeneous under this partition; it is not part of the primary B1–B4 comparison.

Regime C — Dirichlet non-IID severity (controlled sweep). N-BaIoT traffic is repartitioned using a Dirichlet distribution with $\alpha \in \{0.1, 0.3, 0.5, 1.0, 10.0, \text{IID}\}$ across $K = 20$ synthetic clients. Lower α means higher heterogeneity. Regime C provides secondary controlled non-IID severity evidence for the conditions under which threshold personalization is beneficial.

B. Architecture and Training

The AE encoder has dimensions $[d, 80, 40, 20]$ (ReLU activations, no batch normalization), where $d = 115$ for N-BaIoT and $d = 39$ for CICIoT2023. FL simulation uses the Flower framework [4] with FedAvg for up to 150 rounds and a relative-change convergence criterion (window = 10 rounds, tolerance = 0.005); clients perform E=1 local epoch per round with Adam (lr = 10^{-3}, batch size 256). The choice E=1 keeps training dynamics conservative and fixed while isolating post-training threshold scope; evaluating larger E is future work because it changes client drift and reconstruction-error geometry. With E=1 and full client participation, FedAvg approaches synchronous distributed training with frequent aggregation; the FedAvg framing is retained for consistency with FL-AE literature, and the post-training threshold comparison is invariant to this distinction. B0 trains a centralized AE on pooled benign data, applying the pooled 95th-percentile threshold; it is excluded from the B1–B4 controlled ladder.

C. Evaluation Protocol

Five seeds (0–4) are used. The AE is trained once per seed; B1–B4 thresholds are derived from shared per-client score artifacts. The calibration quantile is $q = 0.95$; clients with fewer than $n_{\min} = 100$ benign calibration samples are calibration-pending, receive τ_{global} as fallback, and are excluded from CV(FPR). Budget-capped runs (150 rounds without satisfying the convergence criterion) reuse the final

TABLE II
EXPERIMENTAL REGIME SUMMARY

Regime	Dataset	K	Purpose
A	N-BaIoT	9	Confirmatory (B1 vs B2)
B	CICIoT2023	63	Condition-of-applicability check
C	N-BaIoT (Dirichlet)	20	Non-IID severity sweep (secondary)

TABLE III
REPRODUCIBILITY SPECIFICATION.

Parameter	Value
AE encoder dims	$[d, 80, 40, 20]$
Loss / Optimizer	MSE / Adam (lr 10^{-3})
Batch size	256
Local epochs per round	1
Rounds budget	150 (convergence early-stop)
Convergence criterion	rel. change < 0.005, window 10
Client participation	Full (all clients each round)
Normalization scope	Per-client (StandardScaler)
Seeds	$\{0, 1, 2, 3, 4\}$
Threshold quantile q	0.95
Cal. eligibility $n_{\min}$	100 benign cal. samples
N-BaIoT split ratios	60/20/18 (train/cal/test, gapped)[a]
CICIoT2023 split	70/15/15 benign; attack test-only[b]
CICIoT2023 cap	50,000 total per client (10,000 attack)
Score artifact reuse	Train once; B1–B4 from shared scores

[a] Chronological split with two 1%-wide temporal gaps to prevent leakage. Nominal test fraction = 0.18; gap buffers excluded from all partitions.
[b] CICIoT2023: benign split 70/15/15; attack reserved for test only (`calibration_benign_only=True`).

AE across all B1–B4 derivations, preserving the threshold-only comparison invariant. In Regime A all 5 seeds converged (rounds 51–93); in Regime B all 5 were budget-capped; in Regime C, 21 of 30 seeds converged (9 budget-capped, concentrated at $\alpha \in \{10.0, \text{IID}\}$).

D. Statistical Analysis

The primary statistic is the per-seed delta $\Delta_s = \text{CV(FPR)}_{\text{B1},s} - \text{CV(FPR)}_{\text{B2},s}$. With only five seeds, seed-level descriptive summaries—mean, standard deviation, range, and sign consistency—are the primary reporting mechanism. A bootstrap CI (10,000 resamples of the five seed-level deltas) is reported as a supplementary interval estimate; it excludes zero but should be interpreted cautiously given the small seed count. Coverage ratio $\rho_{\text{cov}} = |\mathcal{K}_{\text{elig}}|/|\mathcal{K}|$ is reported for all conditions.

Table II summarizes the three regimes. Table III provides the full reproducibility specification.

Environment. All experiments ran on a single workstation: AMD Ryzen 5 7600 (6-core), 12 GB RAM, NVIDIA GeForce RTX 5060 Ti (16 GB), Ubuntu 24.04 LTS. Software: Python 3.12.3, PyTorch 2.11.0 (CUDA 13.0), Flower 1.29.0, Ray 2.55.0, NumPy 2.4.4, pandas 2.3.3, scikit-learn 1.8.0. Runtime is not used as an evaluation endpoint in this threshold-scope study.

TABLE IV
REGIME A (N-BAIOT, $K = 9$): PRIMARY CONTROLLED RESULTS.

Baseline	Mean FPR	Mean TPR	CV(FPR)	CV(TPR)	Worst BA	Mean F1	P10 F1
B0 (Centralized)	0.057 ± 0.001	0.829 ± 0.037	0.787 ± 0.031	0.278 ± 0.036	0.653 ± 0.010	0.720 ± 0.054	0.403 ± 0.059
B1 (Client-Averaged)	0.039 ± 0.002	0.738 ± 0.050	$\mathbf{1.017 \pm 0.019}$	0.289 ± 0.029	0.649 ± 0.011	0.563 ± 0.068	0.344 ± 0.062
B2 (Per-Client)	0.043 ± 0.001	0.744 ± 0.019	$\mathbf{0.299 \pm 0.073}$	0.336 ± 0.007	0.651 ± 0.001	0.604 ± 0.042	0.300 ± 0.000
B3 (Family-Mean)	0.051 ± 0.006	0.745 ± 0.018	0.964 ± 0.096	0.336 ± 0.007	0.653 ± 0.004	0.606 ± 0.038	0.299 ± 0.000
B4 (Cluster-Mean)	0.044 ± 0.003	0.718 ± 0.019	0.645 ± 0.045	0.322 ± 0.012	0.654 ± 0.007	0.545 ± 0.036	0.299 ± 0.000

Coverage 9/9, pending 0. Values are mean $\pm$ std over 5 seeds. B0 is a centralized reference and is excluded from the B1–B4 controlled ladder. Worst BA = worst-client balanced accuracy; Mean F1 = mean Macro-F1; P10 F1 = 10th-percentile Macro-F1 across eligible clients.
Primary result: **B1−B2 CV(FPR)** $\boldsymbol{\Delta = +0.718}$, SD $= 0.079$, seed range $[0.588, 0.772]$, all 5 deltas positive; B1−B4 $\Delta = +0.372$, SD $= 0.055$, seed range $[0.319, 0.443]$.

VI. RESULTS AND DISCUSSION

A. RQ1: FPR Dispersion Under the Client-Averaged Shared Threshold

Table IV presents aggregate results for Regime A over 5 seeds. Under B1, CV(FPR) is 1.017 ± 0.019—per-client FPRs vary by more than their mean (a CV above 1 indicates high relative dispersion). The natural N-BaIoT B1 CV(FPR) (1.017) substantially exceeds the IID Dirichlet baseline (0.074), a difference of 0.943, consistent with device heterogeneity contributing to the observed dispersion. B0's CV(FPR) (0.787), although lower than B1's, remains substantially above B2 (0.299), consistent with any single pooled or shared scalar threshold crossing heterogeneous device-error distributions at different exceedance fractions; B0 is included only as a centralized privacy-incompatible reference, not as part of the controlled B1–B4 ladder.

Fig. 3 shows per-client FPR distributions. The worst client (SimpleHome XCS7 security camera) reaches a mean FPR of approximately 0.122 while other clients operate near zero, illustrating the unequal alarm-rate burden imposed by a single shared threshold across heterogeneous devices.

B. RQ2: Held-Out FPR-Dispersion Reduction and Detection-Quality Cost

The primary endpoint is the B1 vs. B2 CV(FPR) delta per seed. Across the five planned seeds, the B1–B2 CV(FPR) deltas are all positive, with mean $\Delta = 0.718$, SD $= 0.079$, and range $[0.588, 0.772]$ (Table VI). These seed summaries are reported descriptively rather than as confidence intervals. A supplementary bootstrap CI (10,000 resamples of the five seed-level deltas) yields $[0.647, 0.769]$ at 95%, excluding zero; given five seeds this is treated as seed-level consistency evidence under the tested protocol.

Per-client calibration (B2) reduces CV(FPR) from 1.017 to 0.299 (coverage: 9/9, pending: 0). B2 reduces CV(FPR) by 0.718 absolute units, corresponding to a relative reduction $(\mathrm{CV}_{B1} - \mathrm{CV}_{B2})/\mathrm{CV}_{B1} = 70.6\%$, achieved with zero additional threshold uplink. Operationally, B2 requires local threshold storage, versioning with the encoder/calibration window, and sufficient benign calibration data; B4 requires transmitting the 4-scalar fingerprint per client.

TABLE V
CLIENT-SPECIFIC SEPARATION DIAGNOSTICS EXPLAINING THE MAIN LOWER-TAIL MACRO-F1 AND FPR EFFECTS (REGIME A, 5-SEED AVERAGES).

Client	Issue	Evidence	Interpretation
SimpleHome XCS7	Highest FPR under B1	Mean FPR ≈ 0.122 (worst client)	Shared τ concentrates false alarms on high-variance devices
Ennio Doorbell	P10 Macro-F1 floor under B2–B4	Lowest AUROC (0.931); 64.9% attack mass below p_{95}	Benign/attack overlap limits detection after personalization

The tradeoff with detection uniformity is clear from Table IV: CV(TPR) rises under B2 (0.336 vs. B1's 0.289), worst-client balanced accuracy is comparable (B1: 0.649, B2: 0.651), and mean Macro-F1 improves (0.604 vs. 0.563). However, P_{10} Macro-F1 falls from 0.344 (B1) to 0.300 (B2). B2 slightly increases mean FPR (0.043 vs. 0.039) but redistributes false alarms more evenly; B1 attains a lower fleet-average FPR partly by concentrating false alarms on high-variance clients. The near-identical P10 Macro-F1 across B2–B4 (≈ 0.300, std < 0.001) reflects a floor effect: the Ennio Doorbell client determines the P10 floor under B2–B4 in all seeds. Table V provides the compact client-level separation diagnostic for the two limiting cases: Ennio Doorbell explains the lower-tail Macro-F1 loss under B2, while SimpleHome XCS7 explains the high-FPR tail under B1.

Local thresholding controls Ennio's benign false alarms at the cost of missing the majority of its low-magnitude attack traffic, explaining the P10 Macro-F1 floor.

Mechanism. Per-client p_{95} thresholding aligns each client's benign tail, mechanically equalizing false-alarm rates on benign test traffic. However, attack reconstruction-error distributions need not shift in parallel with benign distributions: raising the threshold reduces benign false alarms but can swallow low-magnitude anomaly scores, reducing TPR or lower-tail Macro-F1 (Fig. 2 illustrates the benign-side mechanism). B2 should therefore be interpreted as an FPR-dispersion calibration policy, not a uniform detection-quality improvement.

TABLE VI
PER-SEED CV(FPR) AND DELTAS, REGIME A. ALL $\Delta_{\text{B1-B2}}$ VALUES ARE POSITIVE. B4 RECOVERY = (B1 − B4)/(B1 − B2). ROWS ARE PAIRED BY SEED, SO EACH B1–B2 DELTA COMPARES THRESHOLD POLICIES ON THE SAME FROZEN AE AND SCORE ARTIFACTS.

Seed	B1	B2	B4	$\Delta_{\text{B1-B2}}$	B4 Recov. (%)
0	1.045	0.347	0.632	+0.698	59.2
1	1.017	0.251	0.693	+0.765	42.3
2	1.015	0.243	0.654	+0.772	46.6
3	1.018	0.251	0.575	+0.767	57.8
4	0.992	0.404	0.673	+0.588	54.3
Mean	1.017	0.299	0.645	+0.718	51.8

TABLE VII
REGIME A: SHARED-THRESHOLD CONSTRUCTION SENSITIVITY.

Rule	Construction	CV(FPR)	IQR	Max–min FPR	Δ
B1	Arith. mean	1.017±.019	.035±.002	.120±.007	+.718
B1-pool	Pooled p_{95}	0.805±.048	.084±.007	.134±.012	+.506
B1-wt	Wt. mean	0.907±.033	.050±.018	.129±.003	+.608
B2	Per-client p_{95}	0.299±.073	.015±.003	.041±.012	—

Values: mean ± std, 5 seeds (q = 0.95). No AE retraining; B1-pool = pooled p_{95}; B1-wt = sample-weighted mean of local p_{95}. Δ = vs B2. All B1 variants show larger CV(FPR) than B2, consistent with the advantage not being driven solely by arithmetic-mean construction. B1-pooled has lower CV but higher IQR: raising mean FPR reduces relative dispersion while absolute spread remains larger.

All five seed-level deltas are positive (range 0.588–0.772, Table VI).

Absolute FPR dispersion shows the same B1>B2 ordering: IQR(FPR) decreases from 0.035±0.002 (B1) to 0.015±0.003 (B2), and the max–min FPR gap narrows from 0.120 ± 0.007 to 0.041 ± 0.012. Because CV(FPR) can be sensitive when mean FPR is small (mean FPR under B1: 0.039; under B2: 0.043), absolute dispersion measures are reported alongside it throughout. For B1, CV(FPR) = 1.017 and mean FPR = 0.039 imply $\sigma_{\text{FPR}} \approx 0.040$; the CV above 1 therefore reflects genuine cross-client spread comparable to the mean, not only a small-denominator artifact.

Shared-threshold construction sensitivity. Pooled and sample-weighted shared-threshold variants preserve the B2 advantage (Table VII), suggesting the B1-to-B2 reduction is not driven solely by the arithmetic-mean construction.

C. RQ3: Partial Recovery by Cluster-Mean Calibration

Cluster-mean calibration (B4) achieves CV(FPR) = 0.645± 0.045, a partial but consistent improvement over B1. All five B1−B4 deltas are positive (mean Δ = 0.372, SD = 0.055, range [0.319, 0.443]); all five B4−B2 deltas are also positive (mean Δ = 0.346, range [0.269, 0.442]). B4 recovers approximately 52% of B2's improvement (per-seed: 42%–59%, Table VI). B4 requires no device taxonomy and transmits only compact 4-scalar fingerprints per client; however, it does not provide privacy (see Section IV). Cluster assignments show moderate stability across seeds in Regime A (pairwise adjusted

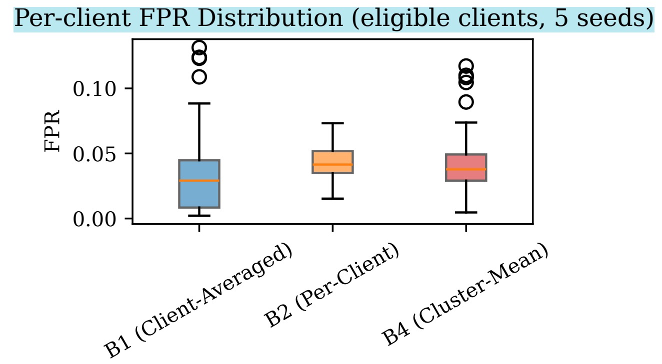

Fig. 3. Per-client FPR distributions under B1, B2, and B4 (Regime A, pooled over 5 seeds). Descriptive only; inference uses seed-level deltas reported in text.

Rand index: mean 0.79, range [0.64, 1.0] over all seed pairs). A post-hoc silhouette check over $K \in \{2, 3, 4, 5\}$ yields mean scores of 0.29, 0.31, 0.34, and 0.32 respectively—all low and within ±0.05—confirming that the nine-device Regime A fingerprint space has no clearly dominant cluster count; K=3 is a pragmatic fixed choice rather than a structurally optimal partition. With only nine physical devices, B4 is exploratory and requires validation on larger fleets. B4's fingerprint still depends on calibration-error statistics including p_{95}, so clients below the eligibility threshold remain calibration-pending and use τ_{global}.

Family-mean calibration (B3, Regime A only) reduces CV(FPR) negligibly (0.964 vs. B1's 1.017): the coarse N-BaIoT device-family taxonomy (Danmini/Ennio doorbells, Ecobee thermostat, Philips baby monitor, Samsung webcam, Provision/SimpleHome security cameras) does not align with the reconstruction-error calibration structure—devices within the same family can have dissimilar benign-error distributions, while devices across families can be similar. This taxonomy-independence of calibration structure motivates B4's data-driven clustering over B3's label-based grouping.

D. Regime B: File-Level Applicability Check (CICIoT2023)

Table VIII shows Regime B results. Under the file-level CICIoT2023 partition, benign reconstruction-error distributions are near-homogeneous (pairwise JS divergence: mean 0.004, median 0.003, max 0.038 over all $\binom{63}{2}$ pairs and 5 seeds); these low pairwise JS values provide the compact calibration-distribution evidence for the Regime B near-homogeneity interpretation. B1 achieves CV(FPR) = 0.099 ± 0.002; B2 increases it to 0.133±0.007 (all five seed-level B1−B2 deltas are negative; mean −0.034, SD 0.006, range [−0.043, −0.026]). When benign distributions are near-homogeneous under this file-level partition, the shared threshold is already a good fit and per-client calibration introduces variance without benefit—no personalization improvement was observed in this setting.

Interpretation caveat. Regime B uses file-defined CICIoT2023 pseudo-clients (not physical devices) with random-

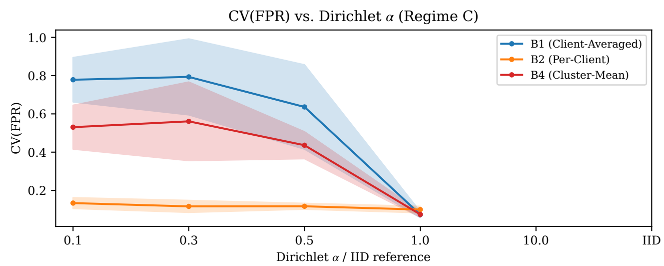

Fig. 4. Regime C: mean CV(FPR) by threshold strategy as a function of Dirichlet α. Whiskers show ± 1 std over 5 seeds. The B1–B2 gap is largest in the low-α band and disappears under IID or near-homogeneous conditions.

ized ordering; it is interpreted only as a near-homogeneous applicability boundary. Intermediate-round score files are not reported; therefore, Regime B is interpreted as a controlled threshold-scope check under a shared final frozen AE, with Macro-F1 stable at approximately 0.959 across threshold strategies, not as a convergence-quality claim.

E. Regime C: Non-IID Severity Sweep

Fig. 4 and Table IX show that the personalization benefit is broadly consistent with larger gains under stronger heterogeneity, concentrated in the low-α band and absent under near-IID conditions. Table IX reports the corresponding seed ranges, so Fig. 4 should be read descriptively rather than as a paired-seed trajectory. The $\alpha = 0.1$ and $\alpha = 0.3$ seed ranges overlap ($[0.535, 0.786]$ vs. $[0.420, 0.897]$), so strict monotonicity between these two levels should not be inferred from this data; they form a high-heterogeneity band rather than a strictly ordered pair. B4 follows the same qualitative pattern but remains a partial approximation.

VII. Threats to Validity

FPR–TPR tradeoff. B2 reduces CV(FPR) but increases CV(TPR) (0.336 vs. 0.289) and decreases P10 Macro-F1 (0.300 vs. 0.344); worst-client balanced accuracy is not materially degraded. In this dataset, the P10 Macro-F1 degradation is concentrated in the low-separability Ennio Doorbell client (benign-vs-attack AUROC 0.931, lowest among all nine clients), so the result is best interpreted as a client-specific failure mode of percentile calibration rather than evidence of a universal FPR–TPR tradeoff. Per-client threshold personalization improves false-alarm dispersion fairness, not universal detection quality.

Dataset scope. Regime A uses 9 physical IoT devices with natural device-level heterogeneity (confirmatory); Regime C ($K{=}20$ synthetic clients) tests severity variation; Regime B provides a contrasting near-homogeneous applicability boundary. The results are a controlled measurement of threshold-scope effects under the tested protocol, not fleet-scale deployment validation; larger multi-vendor fleets are future work.

CICIoT2023 partition. Regime B uses file-level pseudo-clients with randomized within-file benign ordering; no temporal drift is preserved. The null result is therefore an applicability boundary under this randomized near-homogeneous partition only—not evidence that threshold personalization is unnecessary for CICIoT2023 generally. Chronological, physical-device, attack-source, or attack-type CICIoT2023 partitions would define a different heterogeneity regime and remain outside the present threshold-only controlled comparison.

Calibration data sufficiency. In all reported conditions, calibration-pending count is zero. In deployments with fewer than $n_{\min} = 100$ benign calibration samples per client, B2 and B4 fall back to τ_{global}, reducing their advantage.

Threshold quantile sensitivity. q is fixed at 0.95. A sensitivity check over $q \in \{0.90, 0.95, 0.99\}$ preserves the qualitative B2 $<$ B4 $<$ B1 ordering. The B1-pooled and B1-weighted sensitivity checks are consistent with the B1-to-B2 gap not being driven solely by arithmetic-mean construction (Table VII).

Attack scope and adversarial robustness. The evaluation uses Mirai and BASHLITE variants on N-BaIoT. No poisoning, backdoor, or evasion evaluation is performed.

Privacy. B4's fingerprint constitutes distributional metadata leakage; B4 is not a privacy mechanism. Concretely, the mean, variance, skewness, and p_{95} reconstruction-error summaries may reveal device-specific traffic regularity, tail heaviness, or unusual benign-error profiles; secure aggregation or noisy aggregation would reduce this metadata exposure. Mitigations (secure aggregation, differential privacy) are future work.

Convergence and budget-capped runs. Budget-capped runs limit broad convergence interpretation, but within each seed the same frozen AE is used across B1–B4, preserving threshold-calibration-only attribution.

Architecture and $E{=}1$ scope. With $E{=}1$ and full participation, FedAvg approaches synchronous distributed training. The results characterize threshold-scope personalization under a frequently synchronized encoder; they should not be read as evidence that the same magnitude holds under drift-heavy settings ($E > 1$ is future work).

Concept drift. This study uses fixed train/calibration/test splits and does not model online recalibration; DATP assumes benign reconstruction-error distributions remain sufficiently stable between calibration and deployment.

Seeds and statistical power. Five seeds provide limited statistical power. Seed-level descriptive summaries (all deltas positive, range reported) are appropriate for this sample size. All claims are scoped to the tested datasets, partitions, and protocol.

VIII. Conclusion

Under physical-device heterogeneity in N-BaIoT, threshold scope alone strongly changes the distribution of false alarms across clients: per-client p_{95} calibration (B2) reduced CV(FPR) from 1.017 to 0.299 under the fixed FedAvg AE. Cluster-mean calibration (B4) recovers $\approx 52\%$ of B2's improvement without requiring a device taxonomy, at the cost of transmitting distributional metadata that is not private; family-mean calibration (B3) performs poorly because coarse device-type labels do not align with reconstruction-error calibration structure. Under near-homogeneous conditions (Regime B

TABLE VIII
REGIME B (CICIOT2023 FILE-DEFINED CLIENTS, $K = 63$, COVERAGE 63/63, PENDING 0): MEAN $\pm$ STD OVER 5 SEEDS, ELIGIBLE CLIENTS ONLY. B3 NOT APPLICABLE. ALL RUNS BUDGET-CAPPED AT 150 ROUNDS.

Baseline	Mean FPR	Mean TPR	CV(FPR)	CV(TPR)	Worst BA	Mean F1
B0 (Centralized)	0.051 ± 0.001	0.981 ± 0.000	0.109 ± 0.020	0.001 ± 0.000	0.954 ± 0.006	0.959 ± 0.000
B1 (Client-Averaged)	0.051 ± 0.001	0.982 ± 0.000	$\mathbf{0.099 \pm 0.002}$	0.001 ± 0.000	0.959 ± 0.002	0.959 ± 0.000
B2 (Per-Client)	0.051 ± 0.001	0.982 ± 0.000	0.133 ± 0.007	0.001 ± 0.000	0.956 ± 0.001	0.959 ± 0.000
B4 (Cluster-Mean)	0.051 ± 0.001	0.982 ± 0.000	0.101 ± 0.003	0.001 ± 0.000	0.959 ± 0.002	0.959 ± 0.000

Near-homogeneous regime: pairwise JS divergence mean 0.004, max 0.038. B1 achieves lowest CV(FPR); per-client calibration introduces variance without benefit under this partition.

TABLE IX
REGIME C (N-BAIOT DIRICHLET, $K = 20$): CV(FPR) SWEEP.

α	B1 CV	B2 CV	B4 CV	Δ	Seed range
0.1	$.779 \pm .104$	$.134 \pm .025$	$.531 \pm .103$	0.645	[0.535, 0.786]
0.3	$.793 \pm .178$	$.117 \pm .028$	$.561 \pm .184$	0.677	[0.420, 0.897]
0.5	$.636 \pm .197$	$.117 \pm .014$	$.436 \pm .063$	0.519	[0.240, 0.777]
1.0	$.544 \pm .116$	$.096 \pm .021$	$.426 \pm .066$	0.448	[0.238, 0.565]
10.0	$.174 \pm .053$	$.111 \pm .016$	$.133 \pm .012$	0.063	[0.006, 0.122]
IID	$.074 \pm .014$	$.100 \pm .015$	$.075 \pm .013$	-0.026	$[-0.032, -0.019]$

Values are mean CV(FPR) $\pm$ std over 5 seeds. Mean FPR $\approx$ 0.050 for all levels and baselines. B0 excluded. Coverage 20/20 (pending 0) except α=0.1 (19–20/20, pending 0–1). Δ = B1$-$B2 mean delta. Seed range = descriptive min–max over five seeds, not a confidence interval.

file-level CICIoT2023, Dirichlet near-IID) no dispersion reduction is observed—an applicability boundary, not a general recommendation against personalization. B2 is therefore an FPR-equity intervention, not a universal detection-quality improvement: lower-tail P10 Macro-F1 degradation concentrates in the low-separability Ennio Doorbell client. Practitioner guidance is therefore two-step: first measure benign calibration-error heterogeneity, using pairwise distribution divergence or dispersion of local calibration thresholds; then choose the threshold policy according to that heterogeneity and deployment constraints. B1 is appropriate when clients are near-homogeneous or per-client state is undesirable; B2 should be considered when local calibration is available and a shared-threshold pilot shows high relative FPR dispersion, for example CV(FPR) near or above 1 with full calibration coverage; B4 is a taxonomy-free compromise when grouping is acceptable, while accounting for metadata leakage from shared distributional fingerprints.

Limitations and future work. Regime A uses only 9 physical devices (B4 exploratory at this scale); Regime B uses randomized CICIoT2023 file-level pseudo-clients; E=1 with full participation limits local drift; five seeds limit statistical power; no adversarial robustness, formal privacy, or hardware evaluation is provided. Future work covers larger fleets, $E > 1$ sensitivity, privacy-preserving grouping, model-level personalization, adversarial robustness, and deployment validation under concept drift.

Availability. Artifact: Zenodo 10.5281/zenodo.20315654; release v1.0.0; license and reproduction commands are in the repository.

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